

Coupled Critically-Damped Langevin Diffusion for Multi-Task Generative Modeling: Formulation, Analytic Corruption Diagnostics, and Tikhonov-Stabilized Training

1 Setup and notation

We consider N coupled populations, each with state $(X_i, V_i) \in \mathbb{R} \times \mathbb{R}$ (position and an auxiliary velocity variable), stacked as $Z = (X, V) \in \mathbb{R}^{2N}$. Conditioning variables $K_{\text{self},i}$, K_{global} (stiffness references) are exogenous and never depend on (X, V) , which is what makes the whole system *linear* in Z for fixed conditioning and schedule.

Mean-field coupling uses $C_{ij} = 1/(N-1)$ for $i \neq j$, $C_{ii} = 0$, giving each population’s coupling target as the average of the others:

$$\text{target}_i^x = \sum_{j \neq i} C_{ij} X_j, \quad \text{target}_i^v = \sum_{j \neq i} C_{ij} V_j.$$

With time-scale schedule $\tau(t) = t/T$ (the `vp_linear` schedule) and $\hat{\kappa}(\tau) = \tau \cdot K_{\text{global}}$, the per-population drift is

$$\Omega_i^2(\tau) = K_{\text{self},i} + \hat{\kappa}(\tau), \tag{1}$$

$$\begin{aligned} \dot{V}_i = \tau(t) & \left[\alpha_i (-\Omega_i^2(\tau)) X_i + \beta_i \hat{\kappa}(\tau) (\text{target}_i^x - X_i) \right] \\ & + \tau(t) \left[-\gamma_{\text{self}} V_i + \gamma_{\text{couple}} (\text{target}_i^v - V_i) \right], \quad \dot{X}_i = V_i. \end{aligned} \tag{2}$$

β_i is the coupling weight (not friction); $\gamma_{\text{self}}, \gamma_{\text{couple}}$ are the coupled-damping parameters derived in Section 3. Noise enters only through V (hypoelliptic): the diffusion matrix is $G = \sigma I_N$, acting on V alone.

2 Forward and reverse SDEs: the Anderson correction

A naive construction of a “corruption” process negates the entire drift (both the confining and damping terms). This is the standard, *correct* recipe for first-order (DDPM/VP-SDE-style) diffusion models, but is *incorrect* for a second-order (position+velocity) system: negating the damping term converts it from energy-dissipating to energy-*injecting*, producing a genuinely unstable corruption process independent of any schedule or discretization choice.

2.1 Correct derivation via Fokker–Planck matching

For the full state $Z = (X, V)$ with forward SDE $dZ = f(Z, t) dt + \hat{G}(t) dW$ ($\hat{G}$ state-independent, hypoelliptic: $\hat{G} = \begin{pmatrix} 0 \\ G \end{pmatrix}$), matching forward and reverse Fokker–Planck equations gives, for $s = T - t$ (Anderson, 1982):

$$\tilde{f}(z, s) = -f(z, T - s) + \hat{G}(T - s) \hat{G}(T - s)^\top \nabla_z \log p_{T-s}(z).$$

Since $\hat{G} \hat{G}^\top$ has a zero block in X , the score correction only touches the V -equation. Splitting into components:

$$\text{Forward } (t \uparrow): \quad dX = V dt, \quad dV = +f(X, V, t) dt + G dW, \quad (3)$$

$$\text{Reverse } (t \downarrow): \quad dX = -V ds, \quad dV = [-f(X, V, t) + G G^\top \nabla_V \log p_t] ds + G d\bar{W}. \quad (4)$$

The reverse X -update sign flip ($-V ds$, not $+V ds$) is the non-obvious piece: it is required for the deterministic (zero-score) round trip to exactly cancel.

2.2 Verification

For frozen (time-invariant) critically-damped coefficients, the corrected zero-score round trip gives $\Phi = I$ *exactly* (to floating-point precision) for every duration tested. The original (uncorrected) convention gives Φ_{11} growing from $1.05\times$ (short duration) to $19.5\times$ ($S = 1$) to $614\times$ ($S = 2$) — an unbounded exponential instability with closed-form growth rate $\lambda_+ = \sqrt{k}(1 + \sqrt{2})$ for critically-damped k . Under the true time-varying `vp_linear` schedule at the production configuration ($T = 20$, $dt = 0.05$), Φ_{11} dropped from 1.599 (uncorrected) to 1.048 (corrected).

3 Critical damping with mean-field coupling

The mean-field coupling matrix’s eigenvectors are population-independent: the symmetric (all-ones) mode has zero coupling contribution (the target already equals the state when all populations are equal), and the $N - 1$ degenerate antisymmetric modes see the coupling at full relative strength $N/(N - 1)$. This gives exact undamped normal-mode frequencies

$$\Omega_{\text{sym}}^2 = \alpha \Omega^2, \quad \Omega_{\text{anti}}^2 = \Omega_{\text{sym}}^2 + \frac{N}{N - 1} \beta \hat{\kappa},$$

evaluated at the schedule peak $\hat{\kappa} = K_{\text{global}}$. A single scalar γ (population-diagonal damping, as in a naive design) cannot critically damp both modes once $\beta > 0$, since $\Omega_{\text{sym}} \neq \Omega_{\text{anti}}$. The fix couples the damping term the same way the stiffness is coupled (Eq. 2); solving $\Gamma_{\text{sym}} = 2\zeta \Omega_{\text{sym}}$, $\Gamma_{\text{anti}} = 2\zeta \Omega_{\text{anti}}$ for $(\gamma_{\text{self}}, \gamma_{\text{couple}})$:

$$\gamma_{\text{self}} = 2\zeta \sqrt{\Omega_{\text{sym}}^2}, \quad \gamma_{\text{couple}} = \frac{2\zeta(N - 1)}{N} \left(\sqrt{\Omega_{\text{anti}}^2} - \sqrt{\Omega_{\text{sym}}^2} \right).$$

Verified via basis-vector Jacobian extraction to reproduce both target frequencies and both target damping ratios exactly. The $N = 1$ case reduces exactly to standard critically-damped Langevin dynamics ($\gamma_{\text{couple}} = 0$).

4 The analytic transition kernel

Because Eq. 2 is exactly linear in (X, V) for fixed conditioning, the discrete forward transition is an exact linear-Gaussian map. Precomputing, once per configuration,

$$\Phi_{0 \rightarrow t} = M_t M_{t-1} \cdots M_1, \quad \Sigma_t = M_t \Sigma_{t-1} M_t^\top + Q_t, \quad \Sigma_0 = 0,$$

(M_t the per-step Jacobian-derived transition matrix, Q_t the per-step injected-noise covariance) gives the exact conditional kernel

$$Z_t \mid Z_0 \sim \mathcal{N}(\Phi_{0 \rightarrow t} Z_0, \Sigma_t), \quad \nabla_z \log p_t(z \mid z_0) = -\Sigma_t^{-1}(z - \Phi_{0 \rightarrow t} z_0).$$

This was verified against empirical chained-step Monte Carlo simulation (matching to within sampling error, 200,000 samples). Since the ground truth is itself Gaussian ($X_0 \sim \mathcal{N}(0, \Sigma_{\text{gt}})$), the true *marginal* $p_t(z) = \int p_t(z \mid x_0) p(x_0) dx_0$ is also exactly Gaussian:

$$Z_t \sim \mathcal{N}\left(0, \Phi_t \begin{pmatrix} \Sigma_{\text{gt}} & 0 \\ 0 & 0 \end{pmatrix} \Phi_t^\top + \Sigma_t\right),$$

giving a closed-form *exact score oracle* usable with no trained network at all (Section 9).

5 The correlation leak

5.1 Definition

ρ_{leak} is the correlation produced by the round trip (corrupt then reverse) with coupling switched off ($\beta = 0$) and score set to zero — a control isolating spurious correlation from mechanisms other than active coupling.

5.2 Why ρ_{leak} must be minimized

Every diffusion model’s forward corruption must reach a state that is simple and *independently sample-able* at $t = T$: at generation time there is no data to seed from, so sampling begins from a fresh draw of an assumed prior (here, independent per-population noise, $X \sim \prod_i \mathcal{N}(0, \text{prior_std}_i^2)$ — no cross-population correlation is encoded in this draw). If real data correlation has not actually washed out by $t = T$ (i.e. ρ_{leak} is large), the DSM training targets are implicitly built against a $t = T$ marginal that still carries real structure, while the actual prior draw at generation time does not — a train/inference mismatch that exists regardless of network capacity or how well the coupling-drift inductive bias works. This is a separate concern from whether the score network (aided by the coupled drift) can *reconstruct* correlation during the reverse process, starting from that blank prior: minimizing ρ_{leak} ensures the starting point for generation is valid, which is a precondition for that reconstruction to mean anything, not a substitute for it. This is also why the target is $\approx 1\%$ rather than 0% : over-corrupting (Section 9, SNR-leak threshold sweep) degrades realized KL by making the score-estimation task harder, so ρ_{leak} is minimized only down to the point where the prior is clean enough to trust, not further.

5.3 Cause

At $\beta = 0$, symmetric and antisymmetric modes share an identical propagator Φ , but differ in *initial* variance, since X_0 is drawn from the true (correlated) ground truth:

$$v_{\text{sym}} = \sigma_{x, \text{true}}^2 (1 + (N - 1) \rho_{\text{true}}), \quad v_{\text{anti}} = \sigma_{x, \text{true}}^2 (1 - \rho_{\text{true}}).$$

Propagating both through the same Φ and adding identical noise gives $A_{\text{final}} = \Phi_{11}^2 v_{\text{sym}} + \text{Noise}_{\text{final}}$, $B_{\text{final}} = \Phi_{11}^2 v_{\text{anti}} + \text{Noise}_{\text{final}}$. Substituting into the exchangeable-correlation inversion $\rho = (A - B)/(A + (N - 1)B)$ and simplifying (the N factors cancel exactly) gives the closed form

$$\rho_{\text{leak}} = \rho_{\text{true}} \cdot \frac{\text{SNR}_{\text{final}}}{1 + \text{SNR}_{\text{final}}}, \quad \text{SNR}_{\text{final}} \equiv \frac{\text{Signal}_{\text{final}}}{\text{Noise}_{\text{final}}} = \frac{\Phi_{11}^2 \sigma_{x,\text{true}}^2}{\text{Noise}_{\text{final}}}. \quad (5)$$

Verified to 6 significant figures against direct numerical propagation of the full N -population covariance through the round trip.

5.4 Recovering the injected noise from the leak formula

Choosing a target leak fraction (e.g. 1%) fixes a target SNR via $\text{SNR}/(1 + \text{SNR}) = 0.01 \Rightarrow \text{SNR}_{\text{target}} = 0.01/0.99 = 0.010101$. Since $\text{Signal}_{\text{final}}$ is exactly σ -independent (built purely from the drift Jacobian) and $\text{Noise}_{\text{final}} \propto \sigma^2$ exactly (accumulated linearly from $Q_t \propto \sigma^2$), the invariant

$$k \equiv \text{SNR}_{\text{final}}(\sigma) \cdot \sigma^2$$

is independent of which reference σ is used to measure it (verified identical to 6 decimal places across $\sigma_{\text{ref}} \in \{0.05, 0.3, 1.0, 5.0, 50.0\}$, a 1000 $\times$ range). Solving for the calibrated noise scale:

$$\sigma_N = \sqrt{k(N)/\text{SNR}_{\text{target}}}. \quad (6)$$

At the champion $N = 2$ configuration, $k = 0.447419$, giving $\sigma = 6.6554$ for the 1% threshold (compare: $\sigma = 2.905$ for 5%, $\sigma = 21.14$ for 0.1%). This threshold was itself swept (Section 9) and found to trade off non-monotonically against realized KL — 1% is a genuine interior optimum, not an extreme.

Computing $k(N)$: exact matrix-recursion round trip (not simulation)

Φ_{11} in Eq. 5 is the *round-trip* propagator (forward corruption composed with the Anderson-corrected reverse step, score fixed to zero) — the same quantity verified in Section 2: exactly the identity for frozen coefficients, $\Phi_{11} = 1.048$ under the true `vp_linear` schedule at $T = 20, dt = 0.05$. It is *not* a one-way forward map (a one-way forward propagator instead decays substantially over the horizon, and using it here gives a σ_N wrong by roughly 2–4 $\times$). Because Φ_{11} deviates from 1 only through the schedule’s time-dependence, $k(N)$ has no simple algebraic closed form in n_diff_steps or N — but it *is* computable exactly, via a finite composition of matrices, with no simulation, sampling, or training involved anywhere.

Both the forward step (Eq. 3) and the Anderson-corrected reverse step (Eq. 4, score fixed to 0) are exactly linear-Gaussian maps in (X, V) at each step (Section 2), exactly as the forward-only transition kernel of Section 5 already is. The forward step’s exact matrices, M_t^{fwd} (drift) and N_t^{fwd} (noise injection), are given in Section 5’s transition-kernel construction; deriving the reverse step’s matrices directly from its update equations ($V_{\text{new}} = V - dt \cdot (A_{vx}X + A_{vv}V) + \text{noise}$, $X_{\text{new}} = X - V_{\text{new}} dt$) gives

$$M_t^{\text{rev}} = \begin{pmatrix} I + dt^2 A_{vx} & -dt(I - dt A_{vv}) \\ -dt A_{vx} & I - dt A_{vv} \end{pmatrix}, \quad N_t^{\text{rev}} = \begin{pmatrix} -dt\sqrt{dt} G \\ \sqrt{dt} G \end{pmatrix},$$

where A_{vx}, A_{vv} are the same per-step drift Jacobians used in the forward matrix, evaluated at $\beta = 0, \gamma_{\text{couple}} = 0$ (coupling disabled at its single source in the mode frequencies, Section 3,

since $\Omega_{\text{anti}}^2 = \Omega_{\text{sym}}^2$ when $\beta = 0$). The sign flips relative to $M_t^{\text{fwd}}, N_t^{\text{fwd}}$ are exactly the Anderson correction’s X -update sign flip (Section 2), carried through algebraically rather than re-derived. Composing $T = n_diff_steps$ forward steps followed by T reverse steps,

$$\Phi_{\text{rt}} = \left(\prod_{t=T}^1 M_t^{\text{rev}} \right) \left(\prod_{t=T}^1 M_t^{\text{fwd}} \right), \quad \Sigma_{\text{rt}} = (\text{same recursion } \Sigma \leftarrow M\Sigma M^\top + NN^\top, \text{ forward then reverse legs, } \Sigma_{\text{ref}})$$

gives the exact round-trip transition kernel at a chosen reference σ_{ref} , with no stochasticity anywhere — a deterministic sequence of $2N \times 2N$ matrix multiplications, evaluable in well under a second for any N, n_diff_steps of the sizes used in this paper. This was verified two ways: (i) $\Phi_{\text{rt}}[0, 0] = 1.048243$ at $N=2, T=20, dt=0.05$, matching the independently-measured $\Phi_{11} = 1.048$ of Section 2 to four decimal places; (ii) an independent Monte Carlo simulation of the same round trip (real forward/reverse step functions, 2×10^6 samples) agrees with the exact matrix computation to within sampling noise at every N tested.

Taking $\Phi_{\text{rt},x}$ as Φ_{rt} ’s X -block ($N \times N$, the sub-matrix mapping $X_0 \mapsto X_{\text{final}}$ ’s mean, since $V_0 = 0$), the round-tripped X -covariance is $\Phi_{\text{rt},x} \Sigma_{\text{gt}} \Phi_{\text{rt},x}^\top + \Sigma_{\text{rt},x}$, and $\rho_{\text{leak}}(\sigma_{\text{ref}})$ is that matrix’s mean pairwise correlation (exactly, from the matrix — no sampling needed even for this last step). Inverting Eq. 5: $\text{SNR}(\sigma_{\text{ref}}) = \frac{\rho_{\text{leak}}(\sigma_{\text{ref}})/\rho_{\text{true}}}{1 - \rho_{\text{leak}}(\sigma_{\text{ref}})/\rho_{\text{true}}}$, then $k(N) = \text{SNR}(\sigma_{\text{ref}}) \cdot \sigma_{\text{ref}}^2$ (the σ_{ref} -invariant quantity of Eq. 6), and finally σ_N at the chosen leak threshold. Implemented in `synthetic/exact_dsm.py` (`calibrate_sigma_for_leak`, `roundtrip_leak_snr`) and exposed as a standalone tool, `synthetic/calibrate_sigma.py`.

Since $k(N)$ depends on n_diff_steps (through both the discretization of the forward/reverse maps and the schedule’s realized time-dependence at each step), σ_N must be recomputed whenever n_diff_steps changes — `synthetic/anderson_tikhonov_n_sweep.py` now computes σ_N on demand from the current (N, n_diff_steps) via this exact procedure rather than reading a value cached for a different step count, which is what produced an unfair comparison the first time the step count was changed (Section 10.1). Historical note: the originally-reported $N=2, n_diff_steps=20$ value ($k = 0.447419, \sigma = 6.6554$) is not exactly reproduced by this procedure against the current codebase ($k = 0.385, \sigma = 6.175$, a $\approx 7\%$ difference) despite the procedure itself being validated two independent ways (matching the documented Φ_{11} exactly, and Monte Carlo agreement); this is most plausibly attributable to fixes made to `synthetic/` between when that value was first computed and now, rather than an error in this derivation.

6 Hypoellipticity, singular covariance, and ill-conditioned score training

6.1 Exact rank deficiency at $t = 1$

Since $\Sigma_0 = 0, \Sigma_1 = Q_1 = N_{\text{mat}} N_{\text{mat}}^\top$ exactly, where $N_{\text{mat}} \in \mathbb{R}^{2N \times N}$ is the per-step noise-injection matrix (noise enters only V). A $(2N, 2N)$ matrix built from an $(2N, N)$ factor has rank $\leq N$: Σ_1 is *exactly* singular, not merely small, with exactly N zero eigenvalues. Verified via SVD: rank 2 of 4 ($N = 2$), with the two “zero” singular values at $\sim 10^{-8}$ (floating-point noise around an analytically exact zero). By $t = 2, \Sigma_2$ is full rank but still poorly conditioned (smallest singular value 0.0028, condition number 1572).

This is the same structural content that hypoellipticity (Hörmander’s condition) formalizes: a diffusion with degenerate noise starts with a singular reachable-noise covariance and only becomes full rank once the drift ($\dot{X} = V$) has kinematically mixed the V -noise into X over several steps.

6.2 Why the score blows up, and the correct scale

The exact conditional score $\nabla_v \log p_t(x, v | x_0) = -\text{Var}(V | X)^{-1}(v - \mathbb{E}[V | X])$, where $\text{Var}(V | X) = \Sigma_{vv} - \Sigma_{vx}\Sigma_{xx}^{-1}\Sigma_{xv}$ is the Schur complement of Σ_t 's X -block — *not* the marginal Σ_{vv} , which can look deceptively well-conditioned. At $t = 1$: $\Sigma_{vv} \approx 2.2$ (looks fine) but $\text{Var}(V | X) \approx 2 \times 10^{-7}$ (the true source of the blow-up), because X and V are nearly perfectly correlated when X 's entire variance derives from the same single V -noise injection via the kinematic update $X \leftarrow X + V dt$. The ratio $\text{Var}(V | X)/\Sigma_{vv}$ grows monotonically from 0 at $t = 1$ to 0.92 at $t = 20$.

Consequently $\text{Cov}(\nabla_v \log p_t) = \text{Var}(V | X)^{-1}$ exactly (a standard Gaussian precision-matrix identity), so the score's natural scale spans *orders of magnitude* across t — measured directly: implied $|\text{score}|$ scale of $\sim 2 \times 10^9$ at $t = 1$, dropping to ~ 0.31 by $t = 20$. A network trained with unweighted MSE against this target is dominated by the highest-magnitude, hardest-to-represent early- t examples.

7 Tikhonov-regularized DSM

Two structurally different fixes were tested.

7.1 Exact reconstruction (CLD-style, biased-free)

Following Dockhorn et al. (2021)'s Hybrid Score Matching for critically-damped Langevin diffusion, block-Cholesky $L = \text{chol}(\Sigma_t)$ (with X ordered before V) has $L_{vv} = \text{chol}(\text{Var}(V | X))$ automatically, and

$$\text{score}_v = -L_{vv}^{-\top} \varepsilon_v,$$

depending only on ε_v (the V -block of the standard-normal sampling draw), never ε_x — a training target with exact, t -independent unit variance, verified to float precision for $t \geq 2$. This is mathematically exact but **numerically catastrophic in practice**: L_{vv}^{-1} is enormous near $t = 1$ (~ 1581), so any residual network prediction error — inevitable, no network is perfect — is amplified without bound into the reconstructed score, corrupting the last (closest-to-data) reverse step. Result: KL = 454.3, correlation collapsed to -6.6% of true, unchanged by $5\times$ more training (a conditioning failure, not a convergence failure).

7.2 Tikhonov regularization (biased, bounded)

Instead, cap the effective precision:

$$\text{score}_v^{\text{reg}} = -(\text{Var}(V | X) + \lambda I)^{-1}(v - \mathbb{E}[V | X]), \quad (7)$$

a deliberately biased but bounded target (network output *is* the score directly, no reconstruction step). This trades a small, controlled bias near $t = 1$ for numerical stability everywhere. At $\lambda = 0.1$: KL = 0.0101, matching the NDSM baseline (Section 9.4) and beating the exact-reconstruction approach by a factor of $\sim 45,000$.

8 Coupling as an inductive bias, not a learning target

Since Eq. 2 is fully known and linear, the deterministic drift never needs to be learned by the score network — only the score correction (the marginal-vs-conditional gap from Vincent's DSM

theorem) does. This motivates comparing CSHO against DDPM and SDM under matched conditions: DDPM (independent per-population forward corruption, standard ϵ -prediction) and SDM (VP-SDE forward, denoising score matching) both use a single joint SDE with *no* coupling term in the dynamics — any correlation they recover must be learned entirely by the shared network from data. CSHO instead encodes both stiffness coupling (β) and damping coupling (γ_{couple}) directly into the drift (Section 3), giving the score network a structural head start. A controlled test (siloed, non-shared per-population sub-networks, no cross-population network channel) confirmed the drift-level coupling alone produces real, statistically significant correlation ($p = 0.0625$ vs. all zero-coupling controls, every seed positive) — the mechanism is causally real, not an artifact of network capacity.

9 Experiments and ablations

All experiments: coupled-OU ground truth (coupling_strength = 0.6), critically-damped mode-decoupled γ , vp_linear schedule, $T = 20$, $dt = 0.05$, batch size 256, $n = 5$ seeds unless noted. DDPM/SDM baselines share the identical data/coupling/step-count configuration.

9.1 Anderson-SDE correction

Fixing the reverse-SDE sign convention (Section 2) alone, at fixed $\sigma = 0.3$: KL unchanged (0.187 vs. buggy 0.155) but correlation *collapsed* (21.5% vs. buggy 97.1% of true), revealing that the earlier “good” correlation numbers were an artifact of the sign bug’s exponential amplification, not evidence of real coupling recovery.

9.2 SNR-leak threshold sweep

At the corrected dynamics, sweeping the target leak fraction (all $N = 2$, $\beta = 1$, flat σ , 2000 iters):

Leak target	σ	KL	% of true
5%	2.905	0.0125	88.4%
1%	6.655	0.0103	102.9%
0.1%	21.14	0.0381	101.1%

A genuine U-shape: correlation saturates near 1% (diminishing returns, Eq. 5 is bounded), but KL degrades again past it because the scale-mismatch cost of over-corruption (a provably unbounded $\text{KL}_{\text{scale}} = \frac{1}{2}k(c - 1 - \log c)$ term, $c = \text{generated/true variance ratio}$) grows without saturation as σ increases, and the network’s score-estimation task gets harder at high noise. 1% was adopted as the champion threshold.

9.3 Noise schedule shape

At matched round-trip SNR (1%/5% thresholds), flat $\sigma(t)$ beat every growing-schedule alternative tested: linear $\sigma(t) \propto t/T$ ($8\times$ worse KL, half the correlation recovery) and $\sigma(t) \propto \sqrt{t/T}$ ($2.1\times$ worse at 1%, widening to $3.6\times$ worse at 0.1%). Cause: growing schedules concentrate noise in a narrow late window, giving NDSM-style training a far less uniform range of noise levels across the (uniformly-sampled) training distribution than a flat schedule provides.

9.4 DSM training-procedure ablation

All at $N = 2$, champion $\sigma = 6.6554$, 2000 iters:

Training procedure	KL	% of true
NDSM (implicit/Hyvärinen score matching)	0.0103	102.9%
Exact DSM, unweighted	0.0199	102.8%
Exact DSM, marginal-weighted (scalar)	0.0150	103.6%
Exact DSM, Schur-weighted (scalar)	0.0167	102.3%
Exact DSM, Schur-weighted (full matrix)	0.0168	102.4%
CLD-Cholesky (exact reconstruction)	454.3	−6.6%
Tikhonov ($\lambda = 0.1$)	0.0101	99.2%
Exact score oracle (theoretical ceiling)	0.0088	103.5%

At 10,000 iterations and 40,000 evaluation samples (reducing both training and KL-estimator finite-sample noise), the ceiling drops to 0.0065; NDSM reaches 0.0079 and Tikhonov reaches 0.0082 — statistically indistinguishable from each other and from the ceiling, confirming Tikhonov is a genuinely scalable procedure (error shrinks with training budget, unlike CLD-Cholesky, which stays broken regardless of iteration count — a conditioning failure, not a convergence one).

9.5 λ ablation (Tikhonov)

$N = 2$, $\sigma = 6.6554$, 2000 iters, spanning three orders of magnitude:

λ	KL	corr_gen	% of true
0.01	0.0120	0.602	100.5%
0.1	0.0101	0.595	99.2%
1.0	0.1879	0.523	86.7%
10.0	18.29	0.103	17.1%

Clean U-shape: $\lambda = 0.1$ near-optimal; $\lambda = 0.01$ (closer to unregularized) only mildly worse; $\lambda \geq 1$ degrades sharply and monotonically as over-regularization suppresses real signal everywhere, not just near the singularity.

9.6 Step-count robustness sweep

The headline result (Section 10.1) fixes `n_diff_steps = 500`, chosen specifically to make discretization error small enough to trust the comparison. A complementary question is whether CSHO’s advantage over DDPM/SDM holds at more practical, cheaper step counts, or is itself an artifact of that fine discretization. The full $N = 2.5$, $n = 20$ -seed comparison is repeated at `n_diff_steps` $\in \{8, 16, 32, 64, 128\}$, each independently re-calibrating σ_N per Section 5.4 rather than reusing a value computed at a different step count — the exact failure mode diagnosed in Section 10.3, and the reason this sweep is driven by `run_stepcount_sweep.sh` $\rightarrow$ `run_final_synthetic_experiments.sh` $\rightarrow$ `synthetic/anderson_tikhonov_n_sweep.py` rather than a manual per-step-count σ lookup.

Placeholder – run in progress (`run_full_paper_sweep.sh`), restarted after an earlier pass was interrupted by a numerical bug in `sample_and_tikhonov_score_target`: Σ_t at low `t_idx` is exactly rank-deficient (Section 6), so its near-zero eigenvalues sit at floating-point-roundoff level and occasionally went slightly negative once the calibrated σ_N scaled up at $N = 5$, failing the Cholesky factorization the sampling step depends on. Fixed by making the regularizing jitter relative to Σ_t ’s

own magnitude rather than a bare 10^{-8} floor, verified across 10 seeds and against the exact crash parameters with zero further failures. KL, mean across $N = 2..5$:

n_diff_steps	DDPM (KL)	SDM (KL)	CSHO (KL)
8	TBD	TBD	TBD
16	TBD	TBD	TBD
32	TBD	TBD	TBD
64	TBD	TBD	TBD
128	TBD	TBD	TBD

Full per- N , per-seed breakdown and significance tests once complete: `results/experiment_3_synthetic/stepcount` (synthetic/aggregate_stepcount_sweep.py).

10 Headline result

10.1 Integrator error and why the headline run was redone

The exploratory-phase result (Anderson-corrected SDE, mode-decoupled critical damping, σ calibrated per N via Eq. 6 at the 1% leak threshold, $\beta = 1$ fixed, Tikhonov DSM with $\lambda = 0.1$) used $n_diff_steps = 20$ ($dt = 0.05$) for all three methods and $n = 5$ seeds. Two diagnostics run after that result motivated redoing it at a finer discretization before treating it as final:

- **Discretization convergence.** Holding the diffusion horizon $n_diff_steps \cdot dt = 1$ fixed and refining the step count, the champion config’s induced (Φ_T, Σ_T) at 20 steps differs from a well-converged reference (8000 steps) by $\approx 6.7\%$ relative error (first-order convergence, consistent across $N = 2..5$), dropping to $\approx 1.3\%$ at 100 steps and $\approx 0.25\%$ at 500 steps.
- **Score-term dominance.** Comparing the reverse-time drift’s three pieces (Hamiltonian/confinement, Ornstein–Uhlenbeck damping, score correction) along a real sampling trajectory from the trained Tikhonov net, the score term is *not* subdominant the way it is assumed to be in CLD’s own Euler-on-score justification: it is already comparable to the combined deterministic terms at $t = T$ and reaches $\approx 27\times$ their magnitude by $t = 1$, since the deterministic terms are explicitly damped by the $time_scale(t) = t/T$ schedule while the score correction is not. This means both the linear-part discretization error and the score term’s own (still Euler-order, unchanged in this rerun) error are live contributors to any residual bias, and finer discretization is the available lever for the former without a from-scratch integrator redesign.

Given this, the headline comparison was rerun with $n_diff_steps = 500$ ($dt = 0.002$) instead of 20, applied identically to DDPM, SDM, and CSHO so the comparison remains matched-budget, alongside a larger statistical budget ($n = 20$ seeds, 10000 training iterations, 40000 evaluation samples, up from 5/2000/4000) for a properly powered paired Wilcoxon test. A higher-order treatment of the score term itself (the CLD-open-problem direction referenced by Mou et al., 2021) was deliberately *not* pursued for this result given the workshop timeline, and is left for the archival extension.

10.2 Result

$N = 2$ through 5, $n_diff_steps = 500$, $n = 20$ seeds, 10000 training iterations, 40000 evaluation samples. p -values are paired Wilcoxon on $kl_divergence$ (`core/stats.py::compare_configs`),

$n = 20$ matched seeds. DDPM and SDM are as originally run; CSHO is being rerun with σ_N computed via the exact matrix-recursion calibration of Section 5.4 (the first CSHO pass at this step count used a σ_N carried over unchanged from $n_diff_steps = 20$, an invalid calibration – see Section 10.3 below):

N	Method	KL	corr_gen	% of true	p vs. CSHO (KL)
2	DDPM	0.0045	0.600	99.9%	TBD
2	SDM	0.1985	0.831	138.4%	TBD
2	CSHO	TBD	TBD	TBD	—
3	DDPM	0.0090	0.423	99.0%	TBD
3	SDM	0.2959	0.704	164.8%	TBD
3	CSHO	TBD	TBD	TBD	—
4	DDPM	0.0074	0.333	100.1%	TBD
4	SDM	0.3605	0.617	185.9%	TBD
4	CSHO	TBD	TBD	TBD	—
5	DDPM	0.0120	0.272	100.3%	TBD
5	SDM	0.4580	0.538	198.0%	TBD
5	CSHO	TBD	TBD	TBD	—

Source once complete: `results/experiment_3.synthetic/final_champion_seeds20_iters10000_steps500/tikhonov_n_sweep_significance.csv` (`cscho_tikhonov_vs_ddpm/-vs_sdm,kl_divergence` row, per N). σ_N used for this rerun: $N=2$: 5.957, $N=3$: 5.576, $N=4$: 5.399, $N=5$: 5.238 (`python -m synthetic.calibrate_sigma --n-diff-steps 500 --dt 0.002`).

10.3 Analysis: the first CSHO pass at 500 steps used an invalid calibration

The first CSHO run at $n_diff_steps = 500$ (KL 0.0131–0.0359 across N , losing to DDPM at every N with $p \leq 3.9 \times 10^{-4}$) reused σ_N computed at $n_diff_steps = 20$ unchanged. This was flagged as a live risk before that rerun and is now confirmed to matter: σ_N depends on the discretization through $k(N)$ (Section 5.4), and reusing a value calibrated at a different step count does not sit at the intended 1% leak threshold at the new one. CSHO’s KL barely moved between the 20-step exploratory result and the (miscalibrated) 500-step rerun ($N=2$: 0.0101 $\rightarrow$ 0.0131, i.e. no improvement, slightly worse) while DDPM improved by 15–23 $\times$ over the same change – consistent with CSHO’s apparent lack of improvement being an artifact of the invalid calibration masking whatever real gain finer discretization should have given it, rather than evidence CSHO doesn’t benefit from finer discretization at all.

σ_N for this rerun was computed via the exact matrix-recursion round trip (Section 5.4), not simulation or a heuristic correction – verified against the writeup’s own independently-documented $\Phi_{11} = 1.048$ (Section 2) to four decimal places, and cross-checked against an independent Monte Carlo estimate of the same round trip. `synthetic/anderson_tikhonov_n_sweep.py` now computes σ_N from the current (N, n_diff_steps) on every run rather than reading a cached value, so this specific failure mode – a stale calibration silently carried across a step-count change – cannot recur.

This does not yet resolve whether the CSHO-vs-DDPM reversal itself was purely a calibration artifact or partly reflects DDPM genuinely benefiting more from finer discretization (Sec. 10.1’s discretization-convergence diagnostic remains a live, independent contributor, and the two effects were never cleanly separated at the 20-step/20-seed/10000-iteration budget). The corrected rerun

above will show whether CSHO’s KL improves substantially once calibration is no longer confounding the comparison – if it does, the reversal was primarily a calibration artifact; if it doesn’t, discretization-driven DDPM improvement is the more likely explanation and stands on its own.

11 Extension to real coupled-PDE systems

Sections 9–9.6 establish CSHO’s mechanism and its advantage over DDPM/SDM on a closed-form Gaussian coupled-OU ground truth, where every quantity (the true score, the true stationary covariance, the exact transition kernel) is analytically known and independently verifiable. This section tests whether the same coupling mechanism transfers to real, non-Gaussian, spatially-structured coupled-PDE data, using three genuinely coupled multiphysics benchmarks from Multiphysics-Bench (Xie et al., 2025): **TE_heat** (electro-thermal, E-field $\leftrightarrow$ temperature via Joule heating), **VA** (vibro-acoustic, acoustic pressure $\leftrightarrow$ structural stress/displacement), and **E_flow** (electro-fluidic, potential $\rightarrow$ flow – included specifically as a *unidirectional* falsification case: coupling is expected to be neutral here, since flow does not feed back into the driving electric field).

11.1 Setup

Each problem’s output fields are diffused jointly as an N -task CSHO population ($N = 3$ for **TE_heat**/**E_flow**, $N = 12$ for **VA** after splitting complex-valued fields into real/imaginary task pairs), conditioned on the problem’s input field (**mater**, **kappa**, **rho_water** respectively) via a shared CNN backbone. K_{self} , K_{global} (confinement/coupling stiffness) are held fixed rather than learned, mirroring $K_{\text{REFERENCE}}$ in the synthetic experiments – this is what preserves the exact closed-form transition kernel and Tikhonov-DSM machinery of Sections 3–7 unchanged when moving to real, image-like data: `pde/run_experiment.py` calls `synthetic/anderson_sde.py`, `synthetic/exact_dsm.py`, and `synthetic/drift_coupled_gamma.py` directly, not a reimplement, since task coupling lives entirely in Python-list nesting and $(N, N)/(2N, 2N)$ physics matrices, never in the per-task payload tensor’s own shape.

11.2 A physical-validity metric: PDE-residual error

Alongside relative- L_2 closeness to the reference solution, each prediction is also scored by how well it satisfies the problem’s actual governing PDE – a stronger claim than data-fidelity alone, since it tests whether CSHO’s structural coupling produces physically valid fields rather than merely fields that resemble the training distribution. Following DiffusionPDE’s own evaluation convention (Huang et al., 2024), $L_{\text{pde}} = \text{mean}(\text{residual}^2)$ over the spatial grid, computed per governing equation from the problem’s own PDE operator (complex Helmholtz + Joule-heating for **TE_heat**; Poisson potential + flow continuity for **E_flow**; acoustic Helmholtz + structural equilibrium for **VA**), with finite-difference stencils matched to Multiphysics-Bench’s own PINN-loss convention for direct comparability – deliberately *not* that convention’s own arbitrary loss-balancing divisors and boundary/outlier clipping, which are training-stability hacks rather than a physically meaningful metric.

A byproduct of implementing this metric was discovering a genuine bug in Multiphysics-Bench’s own reference code: `identify_mater` (the function locating **TE_heat**’s elliptical material inclusion) reads its own CSV columns – an ellipse’s semi-major axis, semi-minor axis, and rotation angle – as though they were a circle’s center coordinates and radius, misclassifying $\approx 93\%$ of pixels as “inside” the inclusion on the real dataset (confirmed against the benchmark’s own COMSOL geometry-generation script). This implementation uses the geometrically correct rotated-ellipse test instead –

a deliberate, documented departure from otherwise matching the reference implementation exactly, verified to recover the expected inclusion area to within 0.1 percentage points of the analytic ellipse-area formula.

11.3 Score-network architecture ablation

A separate question from whether CSHO’s coupling helps is whether that advantage is an artifact of one particular score-network design. Three architectures are compared, all trained under the identical CSHO/DDPM/SDM SDE machinery: **attention** (a pooled per-task latent vector refined by cross-field attention – the original design); **FNO** (Li et al., 2021, via the **neuraloperator** package); and **SongUNet** (the DDPM++/NCSN++ denoiser from DiffusionPDE’s own EDM codebase, reused directly rather than reimplemented, with its own EDM preconditioning wrappers bypassed in favor of this paper’s Tikhonov-DSM machinery). FNO and SongUNet both operate at native field resolution with no latent bottleneck: each task’s diffused state is the full $(1, H, W)$ field itself, and the exact-transition-kernel machinery of Section 7 applies unchanged via the same flatten-into-rows trick used for the pooled-latent case.

Implementing this ablation surfaced a genuine limitation in the original attention-based network: it conditions the score prediction on the velocity state V alone, never on the position state X , despite the calling code already supplying X at every step – a placeholder gap, not a deliberate design choice. Both new architectures fix this, conditioning jointly on $(X, V, \text{conditioning}, t)$, matching the score dependence a critically-damped Langevin diffusion model is supposed to have (Section 7); the original attention network is left unchanged for this comparison, superseded rather than patched in place.

11.4 Results

Placeholder – run in progress (run_full_paper_sweep.sh), $3 \times 3 \times 3 = 27$ configurations ($\{TE_heat, E_flow, VA\} \times \{csho, ddpm, sdm\} \times \{attention, fno, songunet\}$), $n = 5$ seeds, $n_epochs = 50$, $n_diff_steps = 20$, batch size 64 – grounded in Multiphysics-Bench’s own PINN convergence budget on these exact datasets (20–30 epochs) and DiffusionPDE’s own proven sampling-step count for these exact problems (10 EDM steps), scaled up moderately for CSHO/DDPM/SDM’s stochastic training objective.

Method	Arch	rel_ L_2 (mean)	spectral_ L_2 (mean)	pde_residual (mean)
DDPM	attention	TBD	TBD	TBD
DDPM	FNO	TBD	TBD	TBD
DDPM	SongUNet	TBD	TBD	TBD
SDM	attention	TBD	TBD	TBD
SDM	FNO	TBD	TBD	TBD
SDM	SongUNet	TBD	TBD	TBD
CSHO	attention	TBD	TBD	TBD
CSHO	FNO	TBD	TBD	TBD
CSHO	SongUNet	TBD	TBD	TBD

TE_heat (mean across $\text{Re}\{Ez\}$, $\text{Im}\{Ez\}$, T for rel_ L_2 /spectral_ L_2 ; mean across e_field, heat for pde_residual). Source once complete: results/experiment_2_physics/TE_heat_{attention,fno,songunet}/{cs

Method	Arch	rel_ L_2 (mean)	spectral_ L_2 (mean)	pde_residual (mean)
DDPM	attention	TBD	TBD	TBD
DDPM	FNO	TBD	TBD	TBD
DDPM	SongUNet	TBD	TBD	TBD
SDM	attention	TBD	TBD	TBD
SDM	FNO	TBD	TBD	TBD
SDM	SongUNet	TBD	TBD	TBD
CSHO	attention	TBD	TBD	TBD
CSHO	FNO	TBD	TBD	TBD
CSHO	SongUNet	TBD	TBD	TBD

E_flow (mean across **ec_V**, **u_flow**, **v_flow** for rel_ L_2 /spectral_ L_2 ; mean across **flow_continuity**, **current_continuity** for pde_residual) – the falsification case: coupling is expected to be neutral here (CSHO $\approx$ DDPM/SDM), since **E_flow**’s physics is unidirectional. Source once complete: `results/experiment_2_physics/E_flow_{attention,fno,songunet}/{csho,ddpm,sdm}_results.json`.

Method	Arch	rel_ L_2 (mean)	spectral_ L_2 (mean)	pde_residual (mean)
DDPM	attention	TBD	TBD	TBD
DDPM	FNO	TBD	TBD	TBD
DDPM	SongUNet	TBD	TBD	TBD
SDM	attention	TBD	TBD	TBD
SDM	FNO	TBD	TBD	TBD
SDM	SongUNet	TBD	TBD	TBD
CSHO	attention	TBD	TBD	TBD
CSHO	FNO	TBD	TBD	TBD
CSHO	SongUNet	TBD	TBD	TBD

VA (mean across all 12 real/imaginary task fields for rel_ L_2 /spectral_ L_2 ; mean across all 6 acoustic/structural equations for pde_residual). Source once complete: `results/experiment_2_physics/VA_{attention,fno,songunet}/{csho,ddpm,sdm}_results.json`.

12 Summary

Encoding critically-damped Langevin dynamics with mean-field coupling directly into the diffusion drift — combined with a mathematically correct (Anderson) reverse-time SDE, a noise scale derived from an exact correlation-leak formula and computed via an exact matrix-recursion round trip (Section 5.4), and a Tikhonov-regularized DSM objective that is robust to the covariance singularity induced by hypoellipticity — gives the score network a substantial, statistically robust advantage over learning the joint distribution with an uncoupled continuous-time SDE (SDM), across every population count tested and at both discretization budgets evaluated. Against discrete-time ancestral sampling (DDPM), the first pass at the larger, finer-discretization budget ($n_diff_steps = 500$, $n = 20$ seeds) showed CSHO losing with high statistical confidence, but that pass used a noise scale σ_N invalidly carried over from the $n_diff_steps = 20$ calibration (Section 10.3); a corrected rerun with σ_N recalibrated for $n_diff_steps = 500$ is in progress and will determine whether the coupling mechanism’s advantage over DDPM (established at the exploratory budget, 5–10 $\times$ on KL) survives

the larger budget once calibration is no longer confounding the comparison. The coupling mechanism’s benefit over an uncoupled SDE (SDM) is established at every budget tested; its benefit over discrete-time ancestral sampling (DDPM) at the larger budget is pending this corrected result.

Section 11 extends this comparison from the closed-form synthetic ground truth to three real, non-Gaussian coupled-PDE benchmarks (`TE_heat`, `E_flow`, `VA`), adding a physical-validity metric (PDE-residual error, Section 11.2) alongside data-fidelity, and testing the coupling mechanism’s advantage across three score-network architectures (the original attention design, FNO, and SongUNet, Section 11.3) rather than one – including a fix, surfaced by that ablation, to a placeholder limitation in the original network (conditioning the score on velocity alone, never position). Results for both this architecture ablation and the step-count robustness sweep (Section 9.6) are pending the corresponding runs.